

# AI-Driven Predictive Modelling and Sustainable Interventions for Lake Restoration: A Case Study of Sukhna Lake Using Satellite Data

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**Abstract:** This paper proposes a comprehensive methodology to address eutrophication and ecological degradation in Sukhna Lake, Chandigarh, using AI-driven predictive modelling combined with eco-innovative solutions. By leveraging satellite remote sensing data from Landsat-7 ETM+ and ASTER along with in-situ measurements, this framework enables the prediction of key water quality parameters and recommends sustainable, nature-based interventions. The proposed model forecasts trends in eutrophication using parameters such as Secchi Disk Transparency (SDT), Chlorophyll-a, and Land Surface Temperature (LST), while introducing floating treatment wetlands and bioremediation as adaptive solutions to reduce nutrient inflow. This framework aims to deliver a proactive, scalable approach for urban lake restoration, providing essential insights into sustainable ecosystem management practices globally.

**Keywords:** Predictive Modelling, Remote Sensing, Sustainable Interventions

## 1. Introduction

Urban lakes, such as Sukhna Lake in Chandigarh, are critical resources for biodiversity, water retention, and recreation. However, urban development, nutrient influx, and climate change have intensified eutrophication, sedimentation, and algae growth, significantly degrading lake quality. Despite conservation efforts by local authorities, challenges persist due to unaddressed nutrient inflow, untreated urban runoff, and episodic algal blooms, highlighting the need for data-driven management solutions.

Remote sensing and AI technologies offer innovative avenues to address these challenges. Remote sensing provides real-time, large-scale monitoring of water quality parameters, while AI predictive models facilitate early warning systems and strategic interventions. This study outlines a predictive model and nature-based solutions framework for sustainable lake restoration, focusing on scalable and adaptive measures that align with both ecological and socio-economic objectives.

## **2. Literature Review**

### **2.1. Remote Sensing in Water Quality Monitoring [1]**

Remote sensing offers significant advantages in monitoring large water bodies by analysing satellite-derived spectral reflectance, which provides information on water clarity, algal concentration, and temperature variations. Previous studies have employed Landsat and ASTER data to calculate indices like NDVI (Normalized Difference Vegetation Index), which is crucial in assessing chlorophyll-a concentrations and Secchi Disk Transparency (SDT).

### **2.2. AI-Driven Predictive Models for Environmental Monitoring [2]**

Machine learning models, especially regression-based models and neural networks, have been successfully applied in environmental science to predict water quality parameters based on input data from satellite imagery and in-situ measurements. Studies have shown that integrating AI with remote sensing can improve the prediction of water quality indices like Trophic State Index (TSI) and LST.

## **3. Methodology**

### **3.1. Study Area and Background Information**

Sukhna Lake, located in Chandigarh, India, spans approximately 3 km<sup>2</sup> and is a prominent recreational area and ecological habitat. Its catchment area comprises urban and agricultural zones, resulting in periodic nutrient inflows that drive eutrophication. The lake's degradation stems from factors like urban runoff, untreated sewage, and nutrient overloads, which lead to sedimentation and declining water quality. This study focuses on a predictive model for key water quality indicators and eco-innovative solutions for long-term lake restoration.

### **3.2. Data Collection**

To achieve a detailed understanding of lake health, this methodology incorporates satellite imagery, in-situ measurements, and auxiliary data sources. Each data type is essential for accurately modelling and predicting lake conditions, and potential issues like missing data, equipment failure, and seasonal variability are addressed.

#### **3.2.1. Satellite Image Data [3]**

Satellite data provides spatially consistent information on Sukhna Lake's water quality and temperature.

#### **Primary Satellite Sources:**

Landsat-7 ETM+ and ASTER are selected for their suitability in tracking indices like SDT, Chlorophyll-a, and LST. Landsat-7 ETM+ is particularly useful for spatially distributed water quality analysis, while ASTER provides enhanced detail in the thermal bands.

*Supplementary Data:* Sentinel-2 and MODIS can serve as secondary sources in case of Landsat data gaps, cloud interference, or for capturing higher temporal resolution.

### Parameters Measured:

*Secchi Disk Transparency (SDT)*: Estimated using the blue and red spectral bands, SDT provides insight into water clarity, a critical factor in assessing eutrophication.

*Chlorophyll-a*: Calculated using NDVI and other indices based on red and near-infrared bands. This parameter directly indicates algal growth and nutrient levels.

*Land Surface Temperature (LST)*: Derived from thermal infrared bands, LST reveals temperature variations, contributing to algal bloom and nutrient cycling.

### Temporal and Seasonal Considerations:

*Data Frequency*: Satellite imagery should be collected weekly or bi-monthly across different seasons (dry, monsoon, post-monsoon) to capture seasonal trends in eutrophication.

*Handling Cloud Cover and Data Gaps*: During monsoon season, cloud interference may reduce data availability. For such cases, data interpolation and alternative sources (e.g., Sentinel-2) are recommended to ensure comprehensive temporal coverage.

### 3.2.2.In-Situ Data Collection [4]

Ground-based measurements validate and calibrate satellite data, enhancing model accuracy.

*Sampling Locations and Frequency*:

**Diverse Sampling Points**: Select representative locations across the lake's urban inflow, middle lake, and rural edge to account for spatial variability in water quality.

**Sampling Frequency**: Bimonthly sampling over one year to capture seasonal dynamics and ensure robust data for model training.

*Parameters Collected*:

**Physicochemical**: pH, DO, BOD, COD, Total Phosphorus (TP), and turbidity are essential for assessing nutrient loads, organic content, and pollutant levels.

**Biological Indicators**: Chlorophyll-a and Total Coliform count indicate algal growth and bacterial contamination.

*Quality Control and Calibration*:

**Calibration of Instruments**: Regular calibration is essential to avoid drift in sensor accuracy.

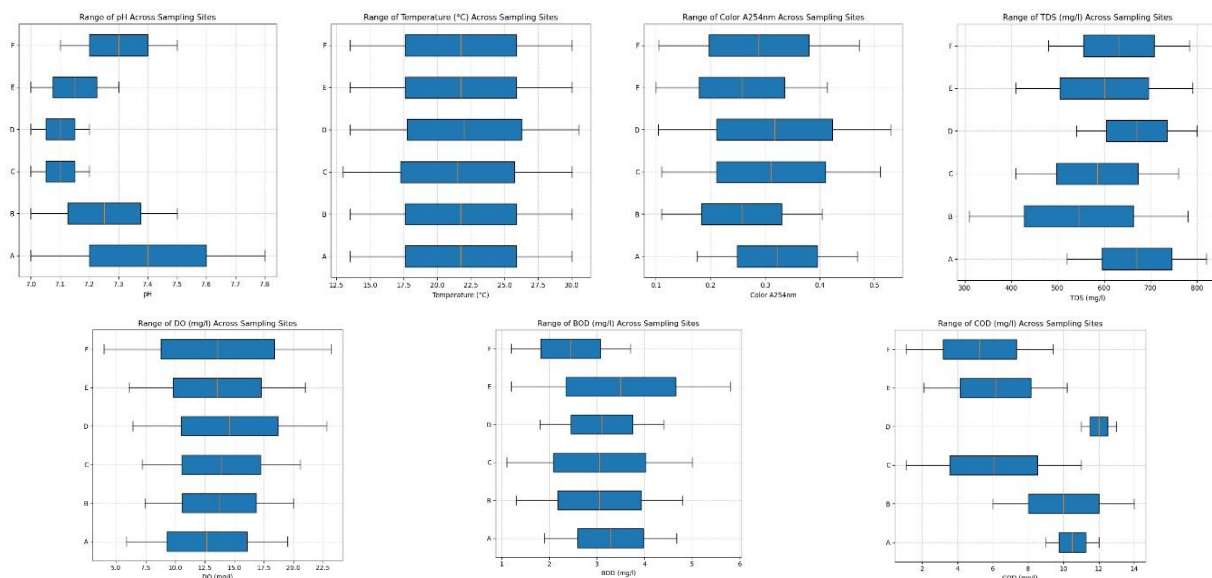
**Validation with Satellite Data**: Ground-based measurements cross-validate satellite-derived parameters, ensuring reliable predictions across varying environmental conditions.

**Table 1:** Results for Physicochemical parameters of Sukhna Lake (Average for Nov-2012, Jan2013, Mar-2013, May-2013, July-2013 and Sept-2013).

| Parameter |         | Site A   | Site B    | Site C     | Site D    | Site E    | Site F    |
|-----------|---------|----------|-----------|------------|-----------|-----------|-----------|
| pH        | Range   | 7.0-7.8  | 7.0-7.5   | 7.0-7.2    | 7.0-7.2   | 7.0-7.3   | 7.1-7.5   |
|           | Mean±SD | 7.2±0.30 | 7.15±0.18 | 7.11±0.075 | 7.06±0.75 | 7.08±0.11 | 7.23±0.17 |
|           | Range   | 13.5-30  | 13.5-30   | 13.0-30.0  | 13.5-30.5 | 13.5-30.0 | 13.5-30.0 |

|              |         |              |             |             |              |               |              |
|--------------|---------|--------------|-------------|-------------|--------------|---------------|--------------|
| Temp (°C)    | Mean±SD | 23.16±6.31   | 23.16±6.31  | 23.16±6.31  | 23.5±6.300   | 23.48±6.31    | 23.41±6.38   |
| Color A254nm | Range   | 0.175-0.469  | 0.110-0.404 | 0.110-0.511 | 0.104-0.53   | 0.100-0.414   | 0.105-0.472  |
|              | Mean±SD | 0.272±0.125  | 0.225±0.09  | 0.205±0.155 | 0.232±0.152  | 0.196±0.113   | 0.195±0.138  |
| TDS (mg/l)   | Range   | 520-820      | 310-780     | 410-760     | 540-800      | 410-790       | 480-783      |
|              | Mean±SD | 693.3±120.56 | 642±172.8   | 655±140.96  | 681.16±99.64 | 642.16±164.87 | 645.5±121.29 |
| DO (mg/l)    | Range   | 5.86-19.5    | 7.46-20.0   | 7.2-20.6    | 6.4-22.8     | 6.1-21.0      | 4.0-23.2     |
|              | Mean±SD | 9.27±5.12    | 10.16±4.86  | 10.8±5.60   | 11.81±7.48   | 11.73±6.25    | 10.93±8.36   |
| BOD (mg/l)   | Range   | 1.9-4.67     | 1.3-4.8     | 1.11-5.0    | 1.8-4.4      | 1.2-5.8       | 1.2-3.7      |
|              | Mean±SD | 3.07±1.17    | 2.68±1.20   | 2.46±1.45   | 2.95±0.88    | 3.64±1.99     | 2.22±0.943   |
| COD (mg/l)   | Range   | 9.0-12.0     | 6.0-14.0    | 1.10-11.0   | 11.0-13.0    | 2.1-10.2      | 1.1-9.4      |
|              | Mean±SD | 9.9±1.1      | 8.0±2.8     | 6.8±4.4     | 7.5±4.4      | 7.6±3.0       | 6.2±3.6      |

SD: Standard deviation



**Figure1.** Box plot showing the range (min to max) for each physicochemical parameter across different sampling sites in Sukhna Lake.

### 3.3. Data Pre-Processing

Data pre-processing ensures the accuracy and usability of satellite and in-situ data for modelling.

### **3.3.1.Satellite Image Pre-Processing**

**Radiometric and Geometric Corrections:** This process converts raw digital numbers (DN) to reflectance values, adjusting for atmospheric and sensor calibration errors. Landsat-7 ETM+ and ASTER data require atmospheric correction using coefficients from metadata to remove influences from atmospheric scattering and absorption.

Images are geometrically aligned to correct distortions due to satellite orientation, Earth curvature, and topographical variations. Ground Control Points (GCPs) are used to map real-world coordinates, ensuring that each pixel corresponds accurately to geographic features on Sukhna Lake.

**Cloud Masking:** Cloud cover often obscures water bodies in satellite images, particularly during monsoon seasons. Automated cloud-masking algorithms (e.g., Fmask) are applied to identify and exclude cloud-covered pixels.

**Handling Missing Data:** Gap-filling techniques such as temporal interpolation are used for gaps created by cloud cover or sensor malfunctions. If data gaps are significant, supplementary satellite sources like Sentinel-2 are utilised for continuity.

**Layer Stacking and Sub setting:** Individual spectral bands are combined to create a multispectral composite image. This enables simultaneous analysis of indices like NDVI and SDT for comprehensive lake quality assessment. The area of interest (AOI) around Sukhna Lake is extracted from the larger image, focusing computational resources and reducing processing time for further analyses.

### **3.3.2.In-Situ Data Processing**

In-situ data complements satellite observations by providing ground-truth measurements. The following steps are essential for ensuring its consistency and reliability:

#### *Data Cleaning:*

**Outlier Detection:** Extreme values caused by measurement errors or anomalous readings are identified and removed using statistical techniques (e.g., z-score thresholds).

**Handling Missing Values:** Missing data points are imputed using temporal averages or interpolation techniques to maintain dataset continuity.

#### *Calibration and Validation:*

**Instrument Calibration:** All equipment, including pH meters, DO probes, and turbidity sensors, are periodically calibrated against standard solutions to prevent drift in accuracy.

**Cross-Validation with Satellite Data:** Parameters such as Secchi Disk Transparency and Chlorophyll-a are compared with corresponding satellite-derived values to calibrate and correct any bias in the model's input.

#### *Normalization and Standardization:*

Unit Standardization: In-situ parameters measured in varied units (e.g., mg/L, NTU) are standardized to directly compare with satellite-derived values. For example, chlorophyll concentrations measured on-site are converted to comparable indices used in remote sensing.

Scaling and Transformation: The data is normalized to ensure that parameters fall within a similar range, facilitating integration into AI models. Scaling methods such as min-max normalization are applied where necessary.

### **3.3.3.Integration of Satellite and In-Situ Data [3]**

The final stage involves merging pre-processed satellite imagery with cleaned and calibrated in-situ data to create a cohesive dataset.

Temporal Synchronization: Satellite imagery and in-situ data are matched based on their collection dates. For instance, satellite images are selected to coincide with in-situ sampling dates, capturing similar environmental conditions.

Spatial Mapping: In-situ sampling points are spatially mapped onto the satellite imagery using GIS software, enabling pixel-to-sample comparisons. This spatial matching enhances the model's ability to detect relationships between satellite-derived parameters and on-site water quality measures.

### **3.4. Predictive Model Development**

The predictive model should accurately forecast eutrophication trends and other water quality parameters.

#### **3.4.1.Model Selection**

##### *Machine Learning Algorithms:*

Random Forest and Neural Networks: Algorithms are trained to predict parameters like Trophic State Index (TSI), SDT, Chlorophyll-a, and LST.

Algorithm Comparison: Test various models, such as Support Vector Machines or Gradient Boosting, to determine optimal prediction accuracy.

GIS Integration: Map nutrient hotspots and eutrophication-prone zones to support seasonal intervention planning.

#### **3.4.2.Model Training and Validation**

Data Splitting: An 80/20 split for training and testing ensures a balanced dataset for model performance evaluation.

Cross-Validation: Perform k-fold cross-validation to test the model's robustness across different sample subsets.

Performance Metrics: To assess model reliability, evaluate accuracy using  $R^2$ , RMSE, and MAE for each parameter.

### 3.5. Proposed Eco-Innovative Interventions

Implementing sustainable solutions such as FTWs and bioremediation addresses nutrient inflow and enhances biodiversity.

#### 3.5.1. Floating Treatment Wetlands (FTWs)

**Design and Placement:** Position FTWs at high nutrient-inflow zones for optimal effectiveness, with adjustments during seasonal changes, especially monsoons.

**Expected Benefits and Maintenance:** FTWs filter excess nutrients, enhance biodiversity, and require regular maintenance for plant health and nutrient absorption efficiency.

#### 3.5.2. Bioremediation

**Microbial Species Selection:** Select microbial species (e.g., algae, bacteria) with proven efficacy in nutrient absorption.

**Application Methods:** Apply bio-augmentation or bio-stimulation techniques as appropriate to boost nutrient removal rates.

**Monitoring Impact:** Continuously measure nutrient levels to assess bioremediation effectiveness.

### 3.6. Contingency and Adaptive Management

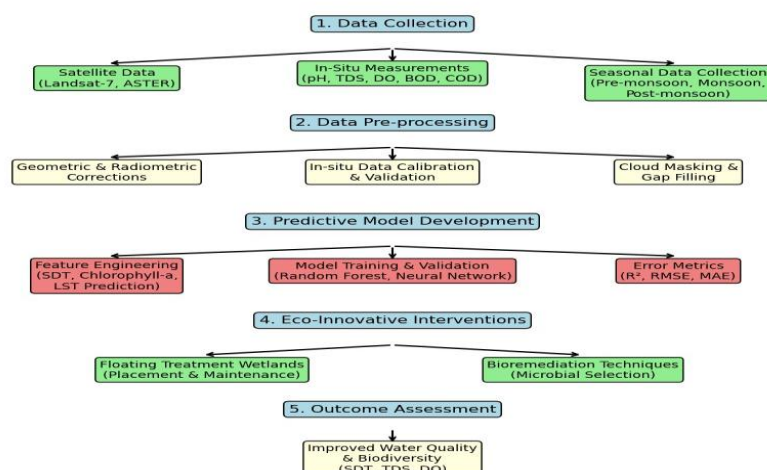
Real-world scenarios, such as unexpected climatic events or equipment issues, demand adaptable management strategies.

**Climatic Adaptation:** Monitor for extreme weather events like droughts or storms that could impact water quality, with adaptive protocols in place.

**Data Gaps and Equipment Failure:** Prepare alternative sources or backup equipment to ensure continuity in data collection.

**Stakeholder Engagement:** Collaborate with local authorities, communities, and environmental organizations to support long-term restoration efforts and foster sustainable lake management practices.

Detailed Flowchart of Lake Restoration Process



**Figure2.** Detailed flowchart of the lake restoration process, illustrating the sequential phases from data collection (including satellite imagery and in-situ measurements) to data preprocessing, predictive model development, and eco-innovative interventions such as Floating Treatment Wetlands and bioremediation. The process culminates in an outcome assessment phase to evaluate water quality and biodiversity improvements, based on key parameters like SDT, TDS, and DO.

#### 4. Expected Outcomes

This proposed methodology is designed to predict eutrophication trends with high accuracy, support proactive interventions, and improve overall lake health through sustainable solutions. The results will highlight the predictive model's ability to guide lake management decisions, demonstrating its potential for wider applicability in urban lake restoration worldwide.

#### 5. Conclusion

The integration of AI-driven predictive modelling with eco-innovative interventions provides a scalable framework for urban lake restoration. This approach enables proactive lake management, promoting sustainability through early intervention and nature-based solutions. Future studies could expand this model to diverse lakes and environments, ensuring adaptable, data-driven lake restoration strategies across regions.

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